

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Database Management Systems
COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO2: Apply the relational model, relational algebra, SQL query structures, and views to solve database querying and data manipulation problems. (BL3, BL6)

Activity Tool : Analytical Problem Solving (High)
Assessment Tool Weightage: 35%

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QUESTIONS			Total Marks: 50
Q.No	Question	Bloom's Level	Marks
1	Apply relational algebra operations (σ , π , $\cup$, $-$, $\times$, $\bowtie$) on the given Employee and Department relations to retrieve required records	BL3 – Apply	5
2	Write SQL DDL statements to create STUDENT(SID, Name, DOB, DeptID) and DEPARTMENT(DeptID, DeptName) tables with all appropriate constraints.	BL3 – Apply	5
3	Formulate SQL queries using GROUP BY, HAVING, and aggregate functions (COUNT, SUM, AVG, MAX, MIN) on a given Sales dataset.	BL3 – Apply	5
4	Write a correlated sub-query to find employees earning more than the average salary of their own department.	BL3 – Apply	5
5	Apply all types of JOIN operations (INNER, LEFT OUTER, RIGHT OUTER, FULL OUTER, CROSS) on Employee and Project tables.	BL3 – Apply	5
6	Create a view 'DeptSalaryView' that displays department name, total employees, and average salary. Write a query on this view	BL6 – Create	5
7	Construct a relational algebra expression for the following: Find names of students who are enrolled in all courses offered.	BL6 – Create	5
8	Write SQL DML statements (INSERT, UPDATE, DELETE) with	BL3 – Apply	5

	integrity constraint enforcement for a given scenario		
9	Apply tuple relational calculus to express a query: Find all employees who work in the 'IT' department and earn more than 50000.	BL3 – Apply	5
10	Design a relational schema for an Airline Reservation System and write 5 SQL queries demonstrating joins, sub-queries, and views.	BL6 – Create	5

Code of Conduct:

I, G Vyshnavi (192525260) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

G Vyshnavi

1. Apply relational algebra operations (σ , π , $\cup$, $-$, $\times$, $\bowtie$) on the given Employee and Department relations to retrieve required records.

```
mysql> SELECT * FROM DEPARTMENT;
+-----+-----+
| DeptID | DeptName |
+-----+-----+
| 101    | CSE      |
| 102    | ECE      |
| 103    | MECH     |
| 104    | EEE      |
+-----+-----+
4 rows in set (0.00 sec)

mysql> SELECT * FROM STUDENT;
+----+-----+-----+-----+
| SID | Name   | DOB       | DeptID |
+----+-----+-----+-----+
| 1   | Rahul  | 2004-05-10 | 101    |
| 2   | Priya  | 2003-08-15 | 102    |
| 3   | Kiran  | 2004-02-20 | 101    |
| 4   | Anjali | 2003-11-25 | 103    |
| 5   | Ravi   | 2004-01-12 | 104    |
+----+-----+-----+-----+
5 rows in set (0.00 sec)
```

2. Write SQL DDL statements to create STUDENT(SID, Name, DOB, DeptID) and DEPARTMENT(DeptID, DeptName) tables with all appropriate constraints.

```
mysql> DESC DEPARTMENT;
```

Field	Type	Null	Key	Default	Extra
DeptID	int	NO	PRI	NULL	
DeptName	varchar(50)	NO		NULL	

2 rows in set (0.05 sec)

```
mysql> DESC STUDENT;
```

Field	Type	Null	Key	Default	Extra
SID	int	NO	PRI	NULL	
Name	varchar(50)	NO		NULL	
DOB	date	NO		NULL	
DeptID	int	YES	MUL	NULL	

4 rows in set (0.00 sec)

3. Formulate SQL queries using GROUP BY, HAVING, and aggregate functions (COUNT, SUM, AVG, MAX, MIN) on a given Sales dataset.

```
Query OK, 7 rows affected (0.05 sec)
```

```
Records: 7 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT * FROM Sales;
```

SaleID	ProductName	Category	Quantity	Price
1	Laptop	Electronics	2	50000.00
2	Mouse	Electronics	5	500.00
3	Keyboard	Electronics	3	1000.00
4	Chair	Furniture	4	3000.00
5	Table	Furniture	2	7000.00
6	Notebook	Stationery	20	50.00
7	Pen	Stationery	50	20.00

7 rows in set (0.00 sec)

4. Write a correlated sub-query to find employees earning more than the average salary of their own department.

```
mysql> SELECT * FROM Employee;
```

EmpID	EmpName	DeptID	Salary
101	Rahul	1	50000
102	Priya	1	60000
103	Kiran	1	45000
104	Anjali	2	55000
105	Ravi	2	65000
106	Sneha	2	50000

```
6 rows in set (0.00 sec)
```

5. Apply all types of JOIN operations (INNER, LEFT OUTER, RIGHT OUTER, FULL OUTER, CROSS) on Employee and Project tables.

```
mysql> DESC Employee;
```

Field	Type	Null	Key	Default	Extra
EmpID	int	NO	PRI	NULL	
EmpName	varchar(50)	YES		NULL	
DeptID	varchar(10)	YES		NULL	
Salary	int	YES		NULL	

```
4 rows in set (0.00 sec)
```

6. Create a view 'DeptSalaryView' that displays department name, total employees, and average salary. Write a query on this view.

```
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> SELECT * FROM DeptSalaryView;
```

DeptName	TotalEmployees	AverageSalary
CSE	2	55000.000000
ECE	2	50000.000000
MECH	1	65000.000000

```
3 rows in set (0.03 sec)
```

7. Construct a relational algebra expression for the following: Find names of students who are enrolled in all courses offered.

```

-> );
+-----+
| Name |
+-----+
| Rahul |
| Kiran |
+-----+
2 rows in set (0.00 sec)

```

8. Write SQL DML statements (INSERT, UPDATE, DELETE) with integrity constraint enforcement for a given scenario.

```

mysql>
mysql> SELECT * FROM Student;
+-----+-----+-----+-----+
| SID | Name | Age | DeptID |
+-----+-----+-----+-----+
| 1 | Rahul | 20 | 101 |
| 2 | Priya | 22 | 102 |
| 3 | Kiran | 19 | 101 |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql>
mysql> DELETE FROM Student
-> WHERE SID = 3;
Query OK, 1 row affected (0.01 sec)

mysql>
mysql> SELECT * FROM Student;
+-----+-----+-----+-----+
| SID | Name | Age | DeptID |
+-----+-----+-----+-----+
| 1 | Rahul | 20 | 101 |
| 2 | Priya | 22 | 102 |
+-----+-----+-----+-----+
2 rows in set (0.00 sec)

```

9. Apply tuple relational calculus to express a query: Find all employees who work in the 'IT' department and earn more than 50000.

```

mysql> SELECT * FROM Employee;
+-----+-----+-----+-----+
| EmpID | EmpName | Department | Salary |
+-----+-----+-----+-----+
| 101 | Rahul | IT | 60000.00 |
| 102 | Priya | HR | 45000.00 |
| 103 | Kiran | IT | 52000.00 |
| 104 | Anjali | Finance | 70000.00 |
| 105 | Ravi | IT | 48000.00 |
+-----+-----+-----+-----+
5 rows in set (0.00 sec)

```

10. Design a relational schema for an Airline Reservation System and write 5 SQL queries demonstrating

joins, sub-queries, and views.

```
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> SELECT * FROM FlightDetails;
+-----+-----+-----+-----+
| PassengerName | FlightName | SeatNo | Fare    |
+-----+-----+-----+-----+
| Rahul         | IndiGo     | A1     | 5000.00 |
| Kiran         | IndiGo     | A2     | 5200.00 |
| Priya         | Air India  | B2     | 6500.00 |
| Anjali        | SpiceJet   | C3     | 7000.00 |
+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```